

Technical Document #129: Software Engineering - Comprehensive Overview

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API design defines how software components communicate. RESTful APIs and GraphQL are popular API design approaches.

Domain-driven design (DDD) models software around business domains. It improves communication between technical and business stakeholders.

Design patterns provide reusable solutions to common software design problems. Singleton, factory, and observer patterns are widely used.

Continuous integration (CI) automatically builds and tests code changes. It catches integration issues early in the development process.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Key Takeaways:

- Technical debt represents the cost of choosing easy solutions over better ones.
- Continuous deployment (CD) automatically deploys code changes to production.
- API design defines how software components communicate.